WORKED SOLUTION

STEP III 1990, Question 6

Pure • Linear Transformations and Ellipses

STEP III 1990, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q6

This question has a full guided treatment in *Beyond STEP*, Section 7.2, *Stretching Without Turning*: https://beyondstep.keithmcnulty.org/ch07/eigenvectors.html

The key idea

The two special vectors are eigenvectors of the matrix, and they tell us the whole geometry: T stretches by one factor along one line and by another factor along a perpendicular line, so the unit circle becomes an ellipse with its axes along those lines. For the algebra, the cleanest route to the equation of E is to invert T, write x and y in terms of X and Y, and substitute into $x^2 + y^2 = 1$. Areas scale by the determinant, and because T maps lines to lines, the tangent to E is simply the image of the tangent to C.

Solution

Write $M = \frac{2}{5}\begin{pmatrix} 9 & -2 \\ -2 & 6 \end{pmatrix}$, so that T is multiplication by M.

The two special directions

We compute

$$M\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \frac{2}{5}\begin{pmatrix} 9-4 \\ -2+12 \end{pmatrix} = \frac{2}{5}\begin{pmatrix} 5 \\ 10 \end{pmatrix} = 2\begin{pmatrix} 1 \\ 2 \end{pmatrix},$$
$$M\begin{pmatrix} 2 \\ -1 \end{pmatrix} = \frac{2}{5}\begin{pmatrix} 18+2 \\ -4-6 \end{pmatrix} = \frac{2}{5}\begin{pmatrix} 20 \\ -10 \end{pmatrix} = 4\begin{pmatrix} 2 \\ -1 \end{pmatrix}.$$

So $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ keeps its direction and is multiplied by 2, and $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ keeps its direction and is multiplied by 4, as required. Notice that these two directions are perpendicular, since $1 \cdot 2 + 2 \cdot (-1) = 0$.

The equation of E

The determinant of M is

$$\det M = \left(\tfrac{2}{5}\right)^2 (9\cdot 6 - (-2)(-2)) = \tfrac{4}{25}\cdot 50 = 8,$$

which is non-zero, so T is invertible and

$$M^{-1} = \frac{1}{8}\cdot\frac{2}{5}\begin{pmatrix} 6 & 2 \\ 2 & 9 \end{pmatrix} = \frac{1}{20}\begin{pmatrix} 6 & 2 \\ 2 & 9 \end{pmatrix}.$$

Hence a point (X, Y) is the image of (x, y) exactly when

$$x = \frac{3X + Y}{10}, \qquad y = \frac{2X + 9Y}{20}.$$

A point (X, Y) lies on E if and only if the point it came from lies on C, that is, if and only if $x^2 + y^2 = 1$. Substituting,

$$\begin{aligned} \frac{(3X+Y)^2}{100} + \frac{(2X+9Y)^2}{400} = 1 &\iff 4(9X^2 + 6XY + Y^2) + (4X^2 + 36XY + 81Y^2) = 400 \\ &\iff 40X^2 + 60XY + 85Y^2 = 400 \\ &\iff 8X^2 + 12XY + 17Y^2 = 80, \end{aligned}$$

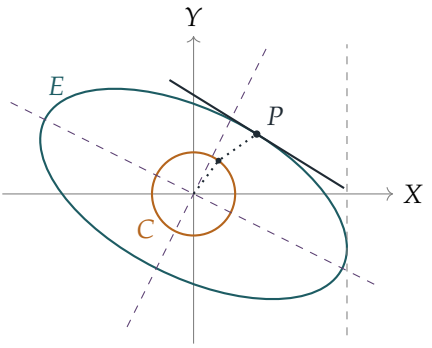
which is the required equation of E.

The area

A linear map multiplies every area by $|\det M|$. The disc bounded by C has area π, so the region bounded by E has area

$$8\pi.$$

This agrees with the geometry: E is an ellipse with semi-axes 2 (along $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$) and 4 (along $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$), and $\pi \cdot 2 \cdot 4 = 8\pi$.



The figure shows C, the ellipse E with its axes (dashed), the vertical line $X = 2\sqrt{17/5}$ touching E, and the tangent at P, the image of $\frac{1}{5}(3, 4)$.

The greatest value of X

Regard the equation of E as a quadratic in Y:

$$17Y^2 + 12XY + (8X^2 - 80) = 0.$$

A given value of X occurs on E if and only if this has a real root Y, that is, if and only if its discriminant is non-negative:

$$\begin{aligned} 144X^2 - 4 \cdot 17(8X^2 - 80) \geq 0 &\iff 144X^2 - 544X^2 + 5440 \geq 0 \\ &\iff 400X^2 \leq 5440 \\ &\iff X^2 \leq \tfrac{68}{5}. \end{aligned}$$

So every point of E has $X \leq \sqrt{68/5} = 2\sqrt{17/5}$, and equality is attained (the discriminant is then zero, giving the single point with $Y = -\frac{6X}{17}$). Hence the greatest value of X on E is $2\sqrt{17/5}$, as required.

Ask yourself Is there a way to see this straight from the circle, without the discriminant?

Yes. On E we have $X = \frac{2}{5}(9x - 2y)$ with $x^2 + y^2 = 1$. Writing $(x, y) = (\cos\theta, \sin\theta)$, the expression $9\cos\theta - 2\sin\theta$ has greatest value $\sqrt{81 + 4} = \sqrt{85}$ (the $R\cos(\theta + \alpha)$ form). So the greatest X is $\frac{2}{5}\sqrt{85} = 2\sqrt{85/25} = 2\sqrt{17/5}$, agreeing with the above.

The tangent

The point $\frac{1}{5}(3, 4)$ on C maps to

$$P = \frac{2}{5}\begin{pmatrix} 9 \cdot \frac{3}{5} - 2 \cdot \frac{4}{5} \\ -2 \cdot \frac{3}{5} + 6 \cdot \frac{4}{5} \end{pmatrix} = \frac{2}{5}\begin{pmatrix} \frac{19}{5} \\ \frac{18}{5} \end{pmatrix} = \begin{pmatrix} \frac{38}{25} \\ \frac{36}{25} \end{pmatrix}.$$

The tangent to C at $\frac{1}{5}(3, 4)$ is perpendicular to the radius there, so it is

$$3x + 4y = 5.$$

Since T is linear and invertible, it maps straight lines to straight lines, and a line meeting C at exactly one point maps to a line meeting E at exactly one point. So the tangent to E at P is the image of this line. Substituting $x = \frac{3X+Y}{10}$ and $y = \frac{2X+9Y}{20}$:

$$\begin{aligned} \frac{3(3X + Y)}{10} + \frac{4(2X + 9Y)}{20} = 5 &\iff 6(3X + Y) + 4(2X + 9Y) = 100 \\ &\iff 26X + 42Y = 100, \end{aligned}$$

so the tangent to E at P is

$$13X + 21Y = 50.$$

As a check, $13 \cdot \frac{38}{25} + 21 \cdot \frac{36}{25} = \frac{494+756}{25} = 50$, so P is on the line. Alternatively, differentiating $8X^2 + 12XY + 17Y^2 = 80$ implicitly, the normal at P is along $(16X + 12Y,\ 12X + 34Y) = \frac{1}{25}(1040, 1680)$, which is parallel to $(13, 21)$, giving the same line.

Checks

- The image of $(1,0)$ is $\frac{2}{5}(9,-2)$, and $\frac{4}{25}\left(8\cdot 81-12\cdot 18+17\cdot 4\right)=\frac{4}{25}\cdot 500=80$, so it lies on E.
- The ends of the axes, $\pm 2\cdot\frac{1}{\sqrt{5}}(1,2)$ and $\pm 4\cdot\frac{1}{\sqrt{5}}(2,-1)$, satisfy the equation of E; for instance $\frac{4}{5}(8+24+68)=80$.
- Numerically, sampling 200 000 points of C gives greatest $X\approx 3.687818$, matching $2\sqrt{17/5}\approx 3.687818$, and the shoelace area of the image polygon is $25.13274\approx 8\pi$.

Common slips

- Applying M rather than M^{-1} when substituting into $x^2+y^2=1$: the equation of the image curve needs the old coordinates in terms of the new ones.
- Forgetting the factor $\frac{2}{5}$ when computing the determinant: it contributes $\left(\frac{2}{5}\right)^2$, not $\frac{2}{5}$.
- Finding the image point P but then using the gradient of C at $\frac{1}{5}(3,4)$ for the tangent to E: linear maps preserve tangency but not slopes.

Beyond STEP

The matrix here is symmetric, and that is why its eigenvectors are perpendicular: this is the two-dimensional case of the spectral theorem, which says every real symmetric matrix can be diagonalised by a rotation. It explains why any quadratic form $ax^2+2hxy+by^2$ describes a conic with perpendicular principal axes, and the eigenvalues of the form's matrix tell you its shape at once. Strang's *Introduction to Linear Algebra* develops eigenvalues, symmetric matrices and quadratic forms with plenty of geometric pictures.

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